



Term Project Stage 1

Team 04

ADA University

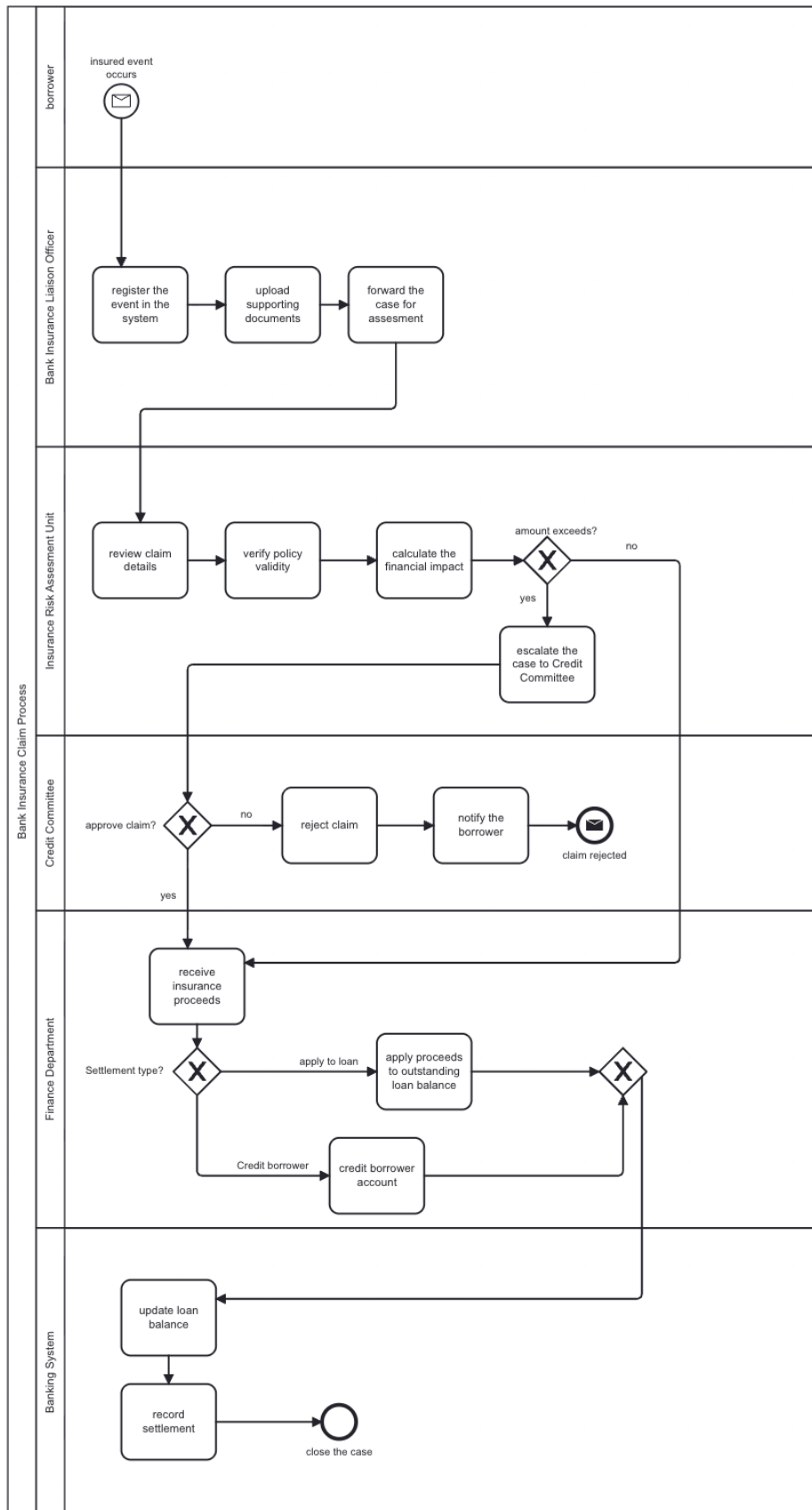
Spring 2026 Business Process Modeling for IT Solutions (INFT-3610 - 20301)

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BPMN Process Model



The Purpose of the Process

The process is used by a bank to ensure the proper management of the insurance claim for mortgaged properties and insure borrowers. As the bank requires some borrowers to take out insurance in accordance with their loan agreements, the bank has a need for a well-defined internal process for reviewing, approving and settling claims when an insured event (for example property damage or life assurance) occurs. This process will enable claims to be thoroughly reviewed. Correct adjustments to the borrower's loan account made, as well as all transactions recorded correctly. In addition, the process provides for larger claims to be approved at the appropriate level, thereby limiting the potential for incorrect settlement of claims.

Process Boundaries

Start event:

When the insured event occurs, it triggers the process to begin, and the bank receives a notification through either the borrower or its insurance partner. Upon receipt of this notice, the bank enters into the process formally.

End events:

There are two possible conclusions to the process. Either the bank will reject the claim, if so, the bank notifies the borrower with the explanation as to why it was denied, and that concludes the process. Alternatively, if the claim is accepted and settled, the process continues until the bank updates the loan account for the borrower, records the settlement in the banking system, and formally closes the file.

Narrative Description of Process

When an event occurs that triggers coverage under an insurance policy held by a customer (such as property damage or death), the customer will report this to their bank. When a customer notifies the bank about an incident, the Bank's Insurance Liaison Officer documents the incident in the bank's system and provides all relevant documentation associated with the reported incident to the Insurance and Risk Assessment Unit for full investigation and assessment of the potential claim.

Upon receipt in the Insurance and Risk Assessment Unit, an assigned officer will review the information contained within the claim in order to determine if the appropriate insurance policy covering the particular insured risk was valid and in force at the time of the event or incident, as well as assess the potential financial effect of the loss on the organization.

Following the determination of the financial impact, the officer will enter the claim information into the system to determine if the proposed claim amount exceeds the bank's predetermined internal authority limits for such claims.

If the claim exceeds the threshold the claim will be escalated to the Credit Committee for review and approval prior to proceeding further. After the Credit Committee reviews the borrower's claim, the committee will either accept or deny the claim. After denying the claim, the banking system notifies the borrower of the denial along with the reason(s) for the denial and the case is closed.

If the borrower's claim does not exceed the threshold or was accepted by the Credit Committee, the case is forwarded to the Finance Department. In this department, the proceeds from the claim are received and how to apply them is determined based upon the type of settlement called for in the agreement.

If the proceeds are to be applied to the loan, the proceeds will be applied directly to the borrower's outstanding loan balance. If the settlement calls for a direct credit, the proceeds will be credited directly to the borrower's account.

After the Banking System applies the proceeds and settles the loan, the Banking System will update the loan balance to reflect the proceeds applied, record the transaction and officially close the case.

List of Roles and Actors and Responsibilities

There are 6 identified roles/actors in this process:

1. Borrower - external client

The client initiates the start of the process by notifying the bank of the event requiring insurance coverage. The borrower's role is to trigger the start of the process, without this trigger process does not start. Later in the process, the borrower gets notified of either approval or rejection of their request.

2. Bank Insurance Liaison Officer - internal employee

Receives the notification from either the borrower or insurance partner, then proceeds with registering the event in the system and uploads all the necessary documents to ensure that the claim case is properly initiated. Once they upload the supporting documents and register the event, they forward the case to the Insurance and Risk Assessment Unit. Without proper registration of the event, downstream assessment cannot take place.

3. Insurance and Risk Assessment Unit - internal organizational unit

Reviews the insurance claim documentation, verifies insurance policy validity and calculates the financial impact. If the amount exceeds the predefined internal

insurance limits, they decide whether the case needs to escalate to the Credit Committee for approval, approves the claim within its authority limits.

4. Credit Committee - internal decision-making authority

Reviews the insurance claims that exceed the internal authority limits, determines whether to approve or reject these claims, then sends a notification to the borrower in case of rejection. If there is no rejection, the process flows to the Finance Department.

5. Finance Department (Officer) - internal employee

Applies insurance proceeds to loan balance, or if needed they credit the borrower account and record the financial transactions. The decision is implemented when they convert approval into records.

6. Bank's System - internal system

The system is responsible for updating the current loan balances and recording the settlement. Once the settlement and updates are stored, the case is closed.

External Entities

External Entities Implied in The Process

1. Insurance Company – external partner

Not depicted in the process diagram, however, once the insured event occurs, they can also initiate the process like the borrower does.

2. Regulatory Authorities – implicit external regulatory body

This could be the Central Bank, Financial Supervisory Authority, or Insurance Regulator. They do not initiate individual claims; however, they define the rules for the processes that take place.

Brief Explanation of Modelling Choices

While modelling, swimlanes were used to depict the entities such as the Borrower, Risk Assessment Unit, and so on. These entities were created inside a single pool named Bank Insurance Claim Process.

Next, tasks were divided into categories based on whether they had human decision-making involved or automated system actions. Activities such as reviewing claim details, verifying policy validity, and calculating financial impact are human made decisions, and activities like updating loan balances and recording the settlement were modelled as system activities.

Then, XOR gateways were used to model a decision-making process, where only one of the two choices had to take place.

To end the processes properly, the circles were used after the activities were executed.

Research

The events that involve insurance of assets that are used as collateral to a loan must follow a structured process that involves both banks and insurance companies. When an insured risk like damage to property takes place, the borrower or any other party involved in the insured risk notifies the risk and provides relevant documentation. The insurance company then evaluates the claim, checks the coverage of the policy and calculates the loss then decides whether to pay the compensation. This practice is typical of the normal insurance claims management that is discussed in risk management books and literature where claims are documented, inquired, assessed and closed based on the terms of the insurance policy [1].

Research on the Azerbaijani insurance industry also indicates that compensation systems are significant towards the management of financial risks and the assurance that the losses incurred due to insured risks can be met by insurers [4].

Once the claim has been evaluated, the banks determine the impact of the insured occurrence

on the loan and the collateral value. The studies on credit risk management in Azerbaijan have shown that banks are applying structured risk evaluation and internal decision processes to examine possible financial effects and make additional steps [2]. In case the financial impact of the claim is high, the decision can be subjected to higher management or credit committee so that there are adequate control and governance [2]. The studies of the mortgage market in Azerbaijan also focus on the fact that the securing of the value of collateral assets is a significant component of the banking system [3]. This is the reason why after insurance claim is accepted, the payment can be used to repair the damaged property or settle the remaining loan balance and then the bank will update the loan records and close the case. These are the practices that show the integration of insured events, verification of claims, risk evaluation, approval of loans and adjustment of loan balances as part of banking operations to guarantee that there is good management of finances [2][3].

References

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Task Breakdown Table

Team Member	Contribution to project	Estimated %
Ulkar Ahmadli	Drew the BPMN model; Purpose of the process; Process boundaries; Narrative description	33.3%
Team leader: Seljan Musayeva	Drew the BPMN model; List of roles and actors, their responsibilities, and external entities; brief explanation of modeling choices	33.3%
Farid Aslanli	Market/academic research; created the research findings section; compiled the citations and the reference list.	33.3%
Lutvu Aliyev	Didn't join groupwork	0%